IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-452

1. Topic (12 words max)

AI-Driven Fabric Defect Detection and Monitoring System for the Apparel Industry in Sri Lanka.

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

The apparel industry is one of the largest contributors to Sri Lanka's economy, providing significant employment opportunities and foreign exchange earnings through exports. However, fabric quality defects remain a persistent challenge, impacting both production efficiency and international competitiveness. Fabric defects such as holes, stains, mis weaves, broken yarns and shade variations are commonly observed during production. These defects, if not identified early, propagate into downstream processes such as cutting and sewing, leading to wastage, increased rework, shipment delays, and ultimately rejected consignments from buyers. Traditional inspection methods in most factories rely heavily on manual inspection tables, where human inspectors visually examine fabric rolls or cut pieces. This method is labor-intensive, time consuming and inconsistent, with detection accuracy depending on inspector experience and fatigue levels. Consequently, defect detection rates remain low, and significant defects often go unnoticed until later stages of garment assembly [1] [2].

Globally, researchers and manufacturers have explored automated quality inspection systems based on machine vision and artificial intelligence to address these limitations. Studies have demonstrated that computer vision, combined with machine learning, can archive superior accuracy and reliability in textile defect detection compared with machine inspectors [3] [4]. For instance, convolutional neural networks (CNNs) and object detection models such as YOLO (You Only Look Once) have been effectively used in real-time defect detection in textile and apparel processes. In parallel, fog and edge computing approaches have been investigated to reduce latency in image processing, making automated inspection feasible in production environments [5] [6]. Additionally, predictive analytics using fusion models that combine machine learning with time-series data have been employed in manufacturing industries to forecast defect trends and optimize production quality [7].

Despite these global advancements, Sri Lanka's apparel industry has yet to fully adopt AI-driven quality monitoring systems. Most local manufacturers continue to rely on manual inspection, which is insufficient for meeting the stringent quality standards of global buyers [8]. The lack of automation in defect monitoring hinders scalability, reduces efficiency, and increases operational costs [9]. Therefore, there is a clear need for a system that can detect fabric defects in real time and provide predictive insights to prevent future occurrences [10]. This research proposes FabricVision, an AI-driven fabric defect monitoring system tailored for the Sri Lankan apparel industry. By integrating high-definition cameras, fog-edge processing, real-time ML models, and predictive fusion analytics, the system aims to revolutionize fabric quality monitoring, ensuring higher consistency, reduced wastage, and enhanced competitiveness in the global market.

References

[1] C. Li, J. Li, Y. Li, L. He, X. Fu, and J. Chen, "Fabric Defect Detection in Textile Manufacturing: A Survey," *Scientific Programming,* 2021.

[2] Carrilho, R.; Yaghoubi, E.; Lindo, J.; Hambarde, K.; Proença, H, "Toward Automated Fabric Defect Detection: A Survey of Recent Computer Vision Approaches," *Electronics,* p. 13, 2024.

[3] Mao M, Hong M., "YOLO Object Detection for Real-Time Fabric Defect Inspection in the Textile Industry: A Review of YOLOv1 to YOLOv11," *Sensors,* vol. 2270, p. 7, 2025.

[4] Li, Y., Song, L., Cai, Y. et al, "Research on fabric surface defect detection algorithm based on improved Yolo_v4," *Sci Rep 14,* vol. 5537, 2024.

[5] Shreshth Tuli, Fatemeh Mirhakimi, Samodha Pallewatta, Syed Zawad, Giuliano Casale, Bahman Javadi, Feng Yan, Rajkumar Buyya, Nicholas R. Jennings, "AI augmented Edge and Fog computing: Trends and challenges," *Journal of Network and Computer Applications,* vol. 216, pp. 1084-8045, 2023.

[6] Resul Das, Muhammad Muhammad Inuwa, "A review on fog computing: Issues, characteristics, challenges, and potential applications," *elematics and Informatics Reports,* vol. 10, pp. 2772-5030, 2023.

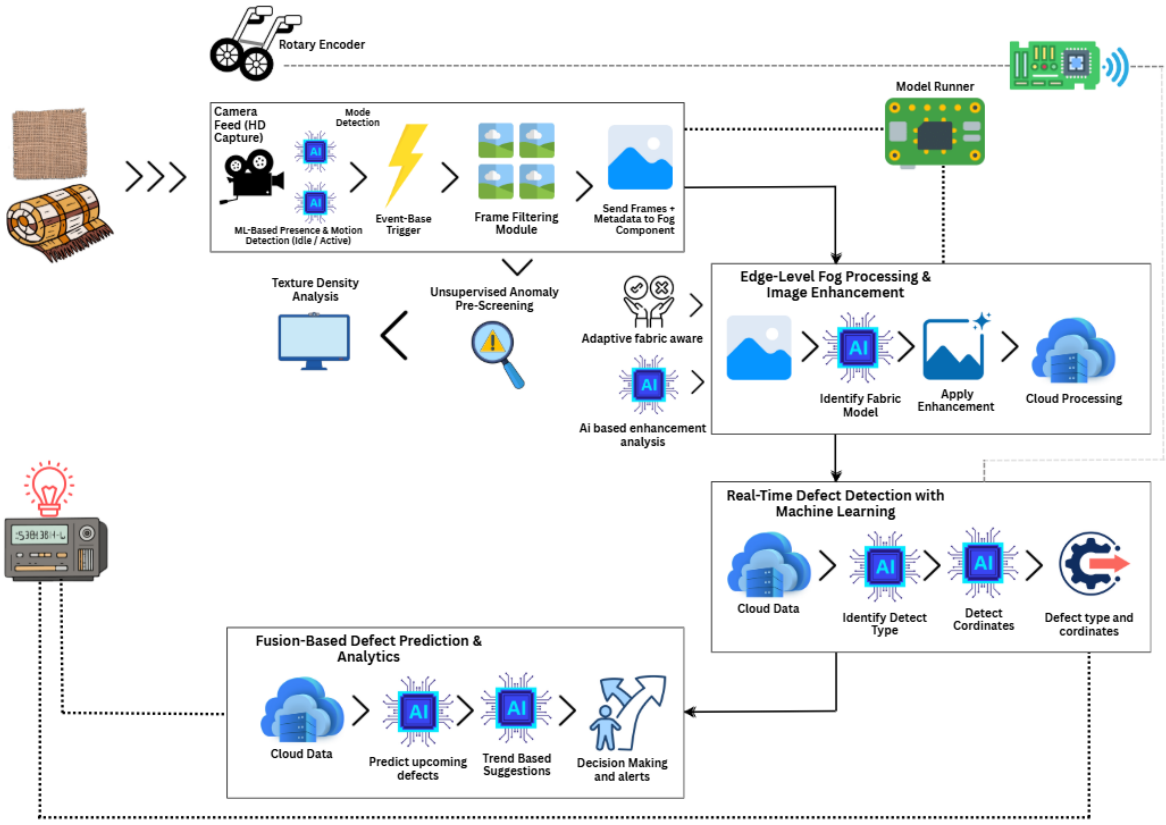
[7] von Enzberg S, Naskos A, Metaxa I, Köchling D and Kühn A, "Implementation and Transfer of Predictive Analytics for Smart Maintenance: A Case Study," *Front. Comput. Sci,* 2020.

[8] P.S.H. Pallemulla, S.J. Sooriyaarachchi, C.R. de Silva and C.D. Gamage, "Defect Detection in Woven Fabrics by Analysis of Co-Occurrence Features," *Engineer: Journal of the Institution of Engineers, Sri Lanka,* 2021.

[9] R. Rathnayake and C. Rajapaksha, "Defect Detection of Knitted Fabrics Using Image Analysis," *roc. SLAAI International Conference,* 2020.

[10] D. P. Kulugammana, "Machine Learning-Based Predictive Model for Fabric Defect Detection," *Proc. ACM,* 2024.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)



All modules are connected to a central decision-making platform, providing operators and supervisors with real-time notifications, dashboards, and reports. Operators receive instant alerts through on-screen messages or physical indicators like lights or buzzers when defects are detected. Supervisors can track defect trends and gain valuable predictive insights through a live dashboard, aiding in decision-making regarding the fabric's quality.

By combining real-time detection with predictive intelligence, the system offers a comprehensive solution for fabric quality monitoring. This system helps reduce waste, minimize human error, improve inspection efficiency, and ensure that garment manufacturers consistently meet international quality standards, ultimately protecting both their profitability and reputation.

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

The apparel industry plays a vital role in Sri Lanka's economy, but ongoing fabric defects remain a major challenge. These defects reduce production efficiency, increase material wastage, and undermine the country's competitiveness in the global market. To address these issues, developing an AI-driven system can offer a cutting-edge solution, combining machine learning, computer vision, and software engineering knowledge to detect and predict fabric defects effectively.

Our team has strong expertise in computer vision, machine learning, and software engineering, which enables us to design advanced defect detection pipelines using technologies like Convolutional Neural Networks (CNNs) and YOLO-based object detection frameworks. This technical knowledge will be supported by collaborations with textile industry operators and quality inspectors, whose insights into fabric structures, common defect patterns, and industry standards will ensure our models are trained to identify the right features and align with quality benchmarks.

In terms of knowledge, the first step is gaining a thorough understanding of textile production processes and the conditions that lead to defects. This includes recognizing defects like holes, stains, misweaves, broken yarns, and shade variations, which may vary depending on fabric type, weave structure, or dyeing methods. Additionally, expertise in fog and edge computing is necessary for real-time image enhancement and low-latency detection at the point of inspection. We'll also need knowledge in predictive analytics to link defect patterns with production data, allowing us to forecast defects and recommend preventive actions. and, a deep understanding of international quality standards and buyer compliance is crucial, as the system's primary goal is to help manufacturers maintain their export competitiveness.

The system will require high-quality datasets to function effectively. This includes quality images of fabric rolls and cut pieces, captured under various lighting, motion, and production conditions. These images must be carefully annotated to distinguish between normal texture variations and actual defects. In addition to image data, we plan to work closely with partner factories, where inspection machines will be equipped with line-scan cameras and IoT sensors to gather real-world data. All data will be systematically labeled, preprocessed, and stored to ensure consistency for training our models.

By combining technical expertise, textile industry knowledge, and ethically collected data, the system aims to remove the gap between manual inspection and automated quality monitoring. The system will be capable of detecting and classifying defects in real-time while predicting future issues. This approach will not only help reduce wastage and improve efficiency but also enhance Sri Lanka's apparel sector's global competitiveness by ensuring high-quality standards are consistently met.

8. Objectives and Novelty

Main Objective:
The objective of this project is to develop an AI-driven real-time fabric defect monitoring system that integrates camera-based inspection, edge-level image enhancement, machine learning–based defect detection, and a centralized dashboard for quality control. The system aims to automate fabric inspection, improve defect detection accuracy, and provide real-time monitoring, defect traceability, and decision support for operators and supervisors.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Nagahawaththa J.C.D IT22573896	Design and develop an intelligent fabric capture and pre-screening module that records only meaningful frames by detecting fabric movement, adapts automatically to roll and cut-piece inspection modes, performs anomaly checking, and generates frame-level and roll-level quality information.	1.1 Calibrate and position the camera for both roll and cut-piece inspections to ensure stable framing, adequate lighting, and consistent fabric visibility. 1.2 Develop ML-based motion & fabric presence detection model to identify the idle vs active states. 1.3 Implement event-based triggering mechanism to capture frames only when meaningful fabric movement or placement is detected.	This component introduces a new event-driven and analytics-focused approach to fabric data acquisition that is not present in existing garment inspection systems. Current solutions rely on continuous frame capturing or static preprocessing, which produces excessive redundant data and lacks early insight into fabric conditions. The proposed module goes beyond these limitations by using ML-based presence detection and event-based triggering to record only meaningful frames, reducing data load while ensuring complete fabric coverage. In addition, the inclusion of an unsupervised anomaly pre-screening

		1.4 Create frame extraction logic and remove blurred, duplicate, or poorly exposed frames while keeping only clear and useful images. 1.5 Adapt the capture pipeline to handle both continuous roll inspection and manual cut-piece placement without requiring manual switching. 1.6 Integrate unsupervised anomaly pre-screening to highlight unusual regions and generate simple risk indicators for each captured frame. 1.7 Generate frame-level and roll-level quality analytics to summarize key metrics such as clarity, stability, and anomaly levels across the inspection session.	process provides early risk indications before the main defect detection stage, something not found in traditional systems that only classify defects after full processing. The component also generates frame-level and roll-level quality analytics such as clarity, exposure stability, and anomaly levels offering operators and downstream modules a richer understanding of fabric condition.
Kamburugamuwa K.S.D IT22574572	Design an edge-level image enhancement and reliability module on a Raspberry Pi device that preprocesses raw fabric	2.1 Retrieve fabric frames from camera feed (Component 1). 2.2 Frame intake and ROI isolation (focus only on fabric area).	Unlike existing fabric inspection systems that rely on fixed preprocessing, this component introduces an ML-driven Adaptive

	frames in real time, ensuring high-quality input for defect detection while improving system stability through health monitoring and adaptive fabric-aware profiles.	2.3 Implement ML-driven Adaptive Fabric-Aware Profile to classify fabric type (light, dark, patterned) and apply optimized enhancement parameters. 2.4 Apply real-time image enhancement (brightness, contrast, denoise, sharpen) using the selected profile. 2.5 Perform AI-based enhancement effectiveness analysis to evaluate improvement in image quality and generate performance metrics for reporting and visualization. 2.6 Implement multi-threaded frame processing to maintain real-time FPS. 2.7 Monitor FPS, latency, and system load to ensure stable performance. 2.8 Trigger health alerts on abnormal conditions (low lighting, noise, overload). 2.9 Forward enhanced frames and analytical metrics to	Fabric-Aware Profile combined with AI-based Enhancement Effectiveness Analysis. An ML model classifies fabric types (light, dark, or patterned) in real time and automatically adjusts enhancement parameters such as brightness, contrast, and sharpening to maintain consistent image quality across varying fabrics. The AI-based analysis then evaluates the improvement in image quality after enhancement and generates performance metrics for reporting and visualization. By combining ML-driven fabric-aware profiling with AI-based analytical evaluation, component becomes an intelligent, analytical, and self-optimizing edge-level enhancement module that ensures high-quality input and reliability for defect detection.

		defect detection (Component 3)	
Kavinda S.G.D IT22603654	Design a predictive and prescriptive analytics module that leverages historical defect data and image-based features to forecast future defect trends, generate risk scores, and provide preventive recommendations and AI report.	3.1 Collect defect records from ongoing fabric rolls and update the database with attributes such as defect type, frequency, position, roll ID, and supplier information. 3.2 Use predictive ML models (XGBoost, LSTM) on the updated data to forecast upcoming defects , patterns and highlight high-risk sections of the roll in advance. 3.3 Generate real-time defect risk scores to the current roll and its supplier, based on predicted defect severity and trends. 3.4 Perform root cause analysis to identify correlations between defects and fabric type, supplier, or operator. 3.5 Produce explainable AI reports that justify assigned	This component goes beyond conventional defect detection by adding prediction, risk evaluation, and explainability to fabric quality control. Unlike existing systems that only log or classify defects, it continuously aggregates inspection data and applies predictive ML (XGBoost, LSTM) to forecast defect trends. It uniquely assigns real-time risk scores for rolls and suppliers, providing quantifiable KPIs for proactive management. The integration of explainable AI reports ensures transparency by justifying predictions and risk levels, building operator trust. By combining forecasting, risk scoring, and explainability, it transforms quality control from a reactive process into a proactive, prescriptive decision-support system not found in current textile QC solutions.

		risk scores, highlighting key patterns and correlations, and present actionable insights for supervisors through the dashboard.	
Jayasinghe J.A.D.T.S IT22028464	Develop a real time defect detection module that combines computer vision and encoder-based tracking using a custom-developed rotary encoder IoT system to detect and localize fabric defects in real time, providing instant alerts, analytical insights and annotated defect images for operator guidance and process control.	4.1 Receive the processed fabric image frames from the cloud storage. 4.2 Use CNN models for texture-based defects (misweaves, broken yarns). 4.3 Use YOLOv8/YOLOv9 for point-based defects (stains, holes, shade variations). 4.4 Develop and implement an IoT-based rotary encoder with measuring wheel to continuously read pulses, convert them to fabric length, and add timestamps. 4.5 Generate annotated images with bounding boxes and labels highlighting detected defects for real-time operator display	Existing fabric quality control systems primarily focus on defect detection and basic classification, often raising alerts without providing precise defect location or contextual information. This system combines computer vision with encoder-based length tracking to not only detect defects in fabric but also pinpoint their exact location in real time. It introduces a novel layer of real-time, granular analytics at the point of inspection. Unlike traditional systems that only raise a defect alert, this solution provides contextual insights (defect type + location) and immediate analytical insights enabling operators to intervene in real-time. The use of CNNs for texture anomalies and YOLOv8/YOLOv9 for point-based issues. It further enhances the system with instant

		4.6 Integrate defect detection with encoder data and send defect + location details instantly. 4.7 Implement IoT-enabled operator feedback using ESP32-based displays or USB/tower lights to indicate OK/REJECT status and provide simplified defect summaries. 4.8 Forword to dashboards showing defect types, locations, and summary statistics to facilitate quick decision making and real time quality control. 4.9 Implement real-time defect analytics, Calculate and stream a live data for current fabric roll based on defect frequency, severity, and type. Generate micro-trend reports (eg, "Stain frequency has increased by 30% in the last 10 meters") and provide immediate feedback on potential machine or material issues.	MQTT updates, annotated defect images, and IoT-enabled displays (ESP32/Tower Lights), providing operators and supervisors with actionable insights immediately. Unlike existing systems, this approach ensures high accuracy across defect types, precise localization, and real time operator guidance

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Nagahawaththa J.C.D IT22573896	The high-definition fabric capture and frame extraction component is built using Python, OpenCV, and lightweight ML models on Raspberry Pi edge device, following strong Software Engineering practices to ensure reliability, scalability, and maintainability. The system is engineered using Agile methodology, modular and layered architecture, structured Git-based version control, CI/CD-driven development, automated testing, API Integration and performance profiling to guarantee stable, real-time operation. The component integrates unsupervised anomaly pre-screening to produce a Visual Irregularity Map (VIM) and Frame Irregularity Score (FIS), along with texture-density analysis using entropy and GLCM features. These additions strengthen data quality, enrich metadata, and improve interoperability with downstream fog and detection modules.
Kamburugamuwa K.S.D IT22574572	This edge-level image enhancement component leverages computer vision techniques to optimize fabric frames for defect detection. Built with Python, OpenCV, and multithreaded pipelines on Raspberry Pi/Jetson devices, it ensures low-latency real-time processing at the edge. Developed under Agile methodology, the system follows modular SE architecture, with clear separation of capture, enhancement, monitoring, and API layers. Integrated health monitoring and adaptive fabric-aware profiles enhance reliability and usability, while SE best practices in performance profiling, scalability, and maintainability ensure the component functions as a robust and operator-friendly edge service.
Kavinda S.G.D IT22603654	The fusion-based defect prediction & analytics component complies with the SE specialization by applying core principles of system design, algorithm development, and decision-support integration to textile quality control. It involves designing a predictive and prescriptive analytics module that not only aggregates and structures defect data but also implements advanced ML algorithms (XGBoost, LSTM) for time-series forecasting. The solution is engineered to assign quantifiable risk scores and integrate explainable AI for transparent decision-making, ensuring reliability and usability for operators and supervisors. By following modular design practices, data-driven modeling, and iterative validation, the component exemplifies software engineering practices that transform complex defect detection outputs into actionable, proactive intelligence for quality management.

Jayasinghe J.A.D.T.S IT22028464	This AI/ML-driven defect detection component integrates advanced computer vision (CNN for texture-based defects, YOLOv8/YOLOv9 for point-based defects) with encoder-based length tracking for precise localization. Developed under Agile methodology, the system follows modular and layered SE architecture, enabling seamless model retraining and updates through CI/CD pipelines. Real-time synchronization between image analysis and encoder data ensures low-latency alerts. The solution emphasize responsible AI deployment by combining real-time industrial safety with SE best practices in performance, scalability, and system resilience.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Thamali	Kelegama	
Co-Supervisor	Ms.	Aparna	Jayawardena	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	✓	No	

b) Does the proposed topic exhibit novelty?

Yes	✓	No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes	✓	No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	✓	No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Accept with minor changes

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	✓
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Added ~~more~~ some functions for 2 components.

Staff Member's Name	Signature
Ms. Thamali Kelegama	Thamali.

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.